Tarek Messaoudi

Full Stack Engineer, Medical Imaging

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Work Experience

Chipiron **Software Engineer** // Medtech building an ultra-low-field MRI scanner for breast imaging **2026 - Present**

🖥️ **MRI Operator Web Application**

- Built the app from scratch, including the exam workflow: drag-and-drop protocols, live scan progress, abort
- Built a JSON Schema-driven form engine for acquisition parameters, with server-side field validation

🩻 **DICOM Imaging Integration**

- Connected Orthanc PACS storage, a reverse proxy with authenticated DICOMweb and a Cornerstone.js viewer

🏥 **RIS/PACS Integration**

- Imported RIS orders as planned exams via DICOM Modality Worklist, with patient data carried into DICOM files
- Sent series to a PACS via C-STORE with per-series retry; added patient search and data erasure

🔐 **Access Control & Quality**

- Implemented authentication and role-based access control in a FastAPI service
- Introduced front-end testing from scratch (Vitest, MSW, Playwright) and OpenAPI-generated API types

Qynapse **Full Stack Engineer** // Healthcare technology company in neuroimaging and AI **2020 - 2025**

🧠 **Medical Imaging Web Platform Redesign**

- Refactored with a new admin panel, interactive data visualizations, and a robust MRI viewer, ...
- Added segmentation correction, lesion tracking, 3D rendering, oblique views and multi-modality

🧬 **Reporting & Domain Tools**

- Rebuilt the report pipeline (Jinja2 + Puppeteer) with templates per language and disease (Dementia, MS, ...)
- Built a normative Python library for imaging metrics and a 3D mesh microservice for structural MRI

📦 **Scalable Infrastructure & Microservices**

- Contributed to converting the academic BrainVisa platform into a Kubernetes-based scalable pipeline
- Expanded the customer-deployed DIMSE agent with routing logic and detailed logging; built an HL7 v2 client

Alter Solutions **Full Stack Engineer** // A consulting and IT services company **2017 - 2020**

- IRT System X: built a central dashboard for research project web services from scratch
- SFR Altice: refactored the customer-service CRM with a new architecture and Mail/SMS notifications
- Thalès: built a local web app to visualize military inventory data from Excel files
- PSA: built a backend service to parse raw automotive files (.a2l, .hex, .ulp)

Technical Skills

Programming Languages: TypeScript, JavaScript, Python, Java
Frontend Development: React, TanStack, Tailwind CSS, Remix
Backend Development: NodeJs, Deno, FastApi, Spring
Databases: PostgreSQL, SQLite, MongoDB, REDIS
Other Technologies: Docker, Kubernetes, Git, Linux, RabbitMQ

Education

IMT Lille-Douai **Software engineer major** // incl. a semester at Chungnam National Univ., South Korea **2012 - 2017**

Projects & Rewards

LOM (GitHub) - *Browser strategy game*

"Entreprise du Futur" Reward (Site) - *By Thalès, from Challenge Entreprendre*